Broker capacity: paired experiment

1K, 5K and 10K | proposal v1 | 3 November 2026

Prepared only - no broker provisioning or load run authorized by this plan

12
matched A/B pairs per class

72
adaptive boundary slots total

24
pre/post drift checks total

Question and first contrast

Estimate a **broker-attributed sustainable publish-rate interval** for one frozen broker and workload. Record a separate low-queue fast-path threshold only when native broker path evidence supports it.

Reference: SMF/TLS Persistent, 1,024-byte seeded payload, two distinct queue copies, four publisher connections, one consumer per queue, exclusive queues.

A/B: subscriber ACK window **255 vs 64** (**-asw**). This first test checks whether consumer flow control explains earlier queue bursts; it is not yet a general broker-feature multiplier.

Randomization and run length

Per class: four sessions across at least two dates, three adjacent A/B pairs per session. The seeded schedule balances **6 AB and 6 BA** orders, with both orders in every session. A safe reference check brackets each session.

Each arm searches a boundary: **60-second scouts**, then independently started **300-second stable and upper trials**. Repeat an upper candidate for another 300 seconds. Search until the decision bracket is at most **1%** wide.

Each of the 72 boundary slots may need several rate trials. The minimum is 12 broker-hours of 300-second confirmations; scouts, drains and repeats add time.

Frozen environment

Prefer Solace PubSub+ **10.26** for all classes. The previous dedicated 5K preflight reported 10.26; revalidate live. Freeze exact build, service ID, VPN, region/AZ, limits, queues, SDKPerf binary hash, Java version, AMI and payload seed.

Candidate generator profile: separate publisher and subscriber **AWS c7i.xlarge** VMs in us-east-1 near the broker. Qualify CPU, NIC/PPS and TCP headroom. If a stronger VM is needed, start a new apparatus epoch and rerun matched arms.

Measurement and rejection gates

Accept a stable trial only if accepted ingress tracks the target by at least **98%**, variation/trend is within **5%**, each destination reconciles unique delivery and CRC, and low-water queue backlog does not trend upward.

An overload candidate needs same-destination egress deficit (at least **2% or 100 msg/s**) plus rising lag and queue floor. Repeat it with independently qualified clients. Client, network, quota or telemetry limits are **inconclusive**, not broker capacity.

For a displayed rate rounded to 100 msg/s, require a validated bracket no wider than **100 msg/s**; otherwise show the interval.

What counts as a meaningful effect?

Analyze paired **log(B/A)** capacity ratios, using conservative boundary endpoints and the wider of pair-level and session-level t intervals. The preregistered smallest effect of interest is **5%**.

Equivalent: full 90% interval inside -5% to +5%. **Material:** full 95% interval beyond +5% or -5%. Otherwise unresolved; add a new 12-pair batch or repair measurement. One standard deviation of raw runs is not an equivalence rule.

A decision applies only to the measured class, broker instance and reference workload. Retest changed payload/fanout contexts and an independent broker before publishing a portable scaling law.

Reproduce and update

Freeze the seeded plan and its SHA-256, raw SEMP/SDKPerf/host traces, exact commands, cost journal, verified boundary artifacts and analysis version. Do not overwrite a finished epoch.

A reviewed result updates the versioned **model/calibration-v1.json** through a new commit and site build. New features need a name, documentation link, applicability and evidence; unverified coefficients stay illustrative.

Readiness: dedicated isolated 1K and 10K targets still needed; 5K requires live revalidation.

Budget: $20/day new generator/traffic ceiling, $16 operating guard; broker subscriptions separate.

Design basis: NIST randomized blocks and practical-equivalence testing; see methodology document for gates.